

mEOL: Training-Free Instruction-Guided Multimodal Embedder for Vector Graphics and Image Retrieval

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TL;DR Revealing the power of structure: training-free multimodal retrieval that finally understands SVG structure

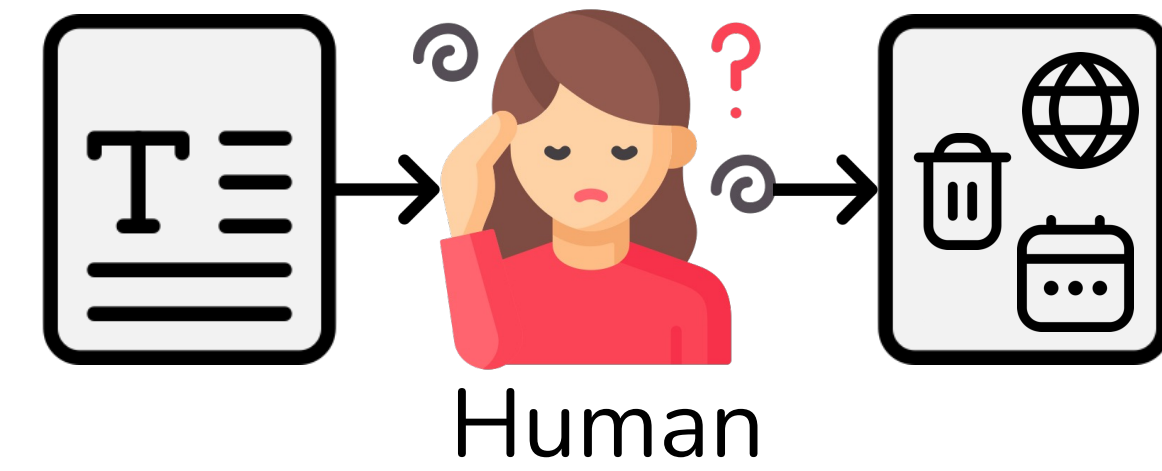
Special thanks to Youwang and GeonU from AMILab.

Problem Statement

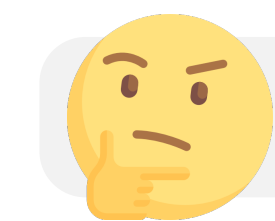
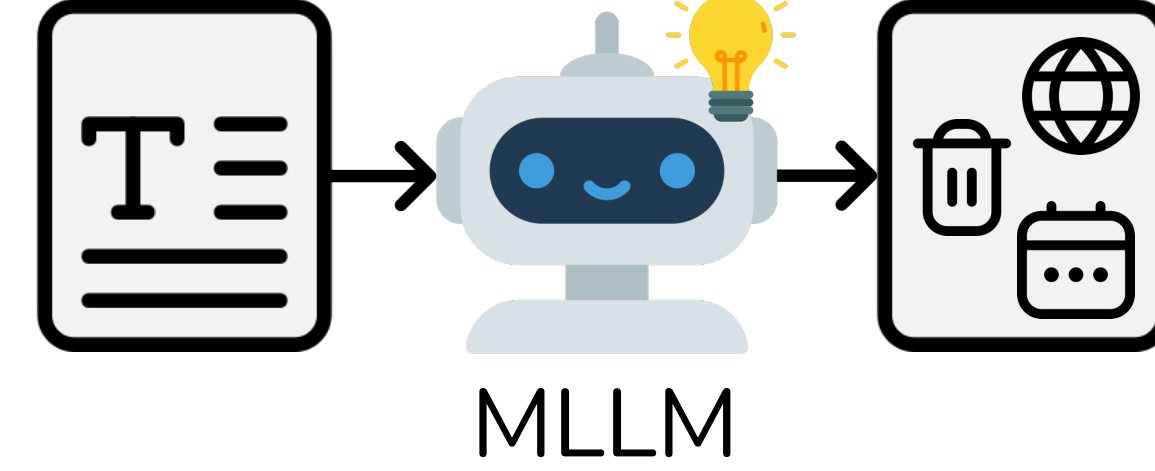
Searching trends moved on from keywords to intent

- For graphic, manually searching takes too long
- Users expect natural queries, not just keywords
- Text-to-SVG retrieval remains unexplored

Traditional methods

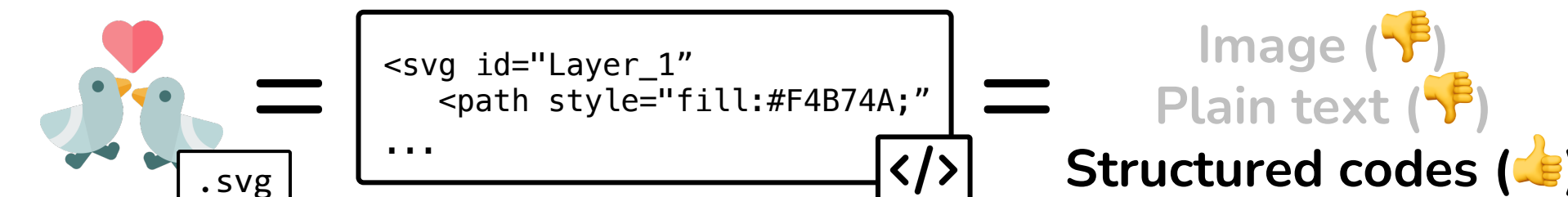


Ours (proposed)



Why Can't We *Just Type* and *Get Perfect icons*?

Preliminaries: SVG & EOL



- A **Scalable Vector Graphic (SVG)** visually resembles an image, but is internally represented as hierarchical vector code

	Text	Image	SVG
Encodes	semantic intent	visual appearance	appearance + structure
Example	Two birds touching...		 + <code></></code>
complexity	compact	fixed-grid	verbose

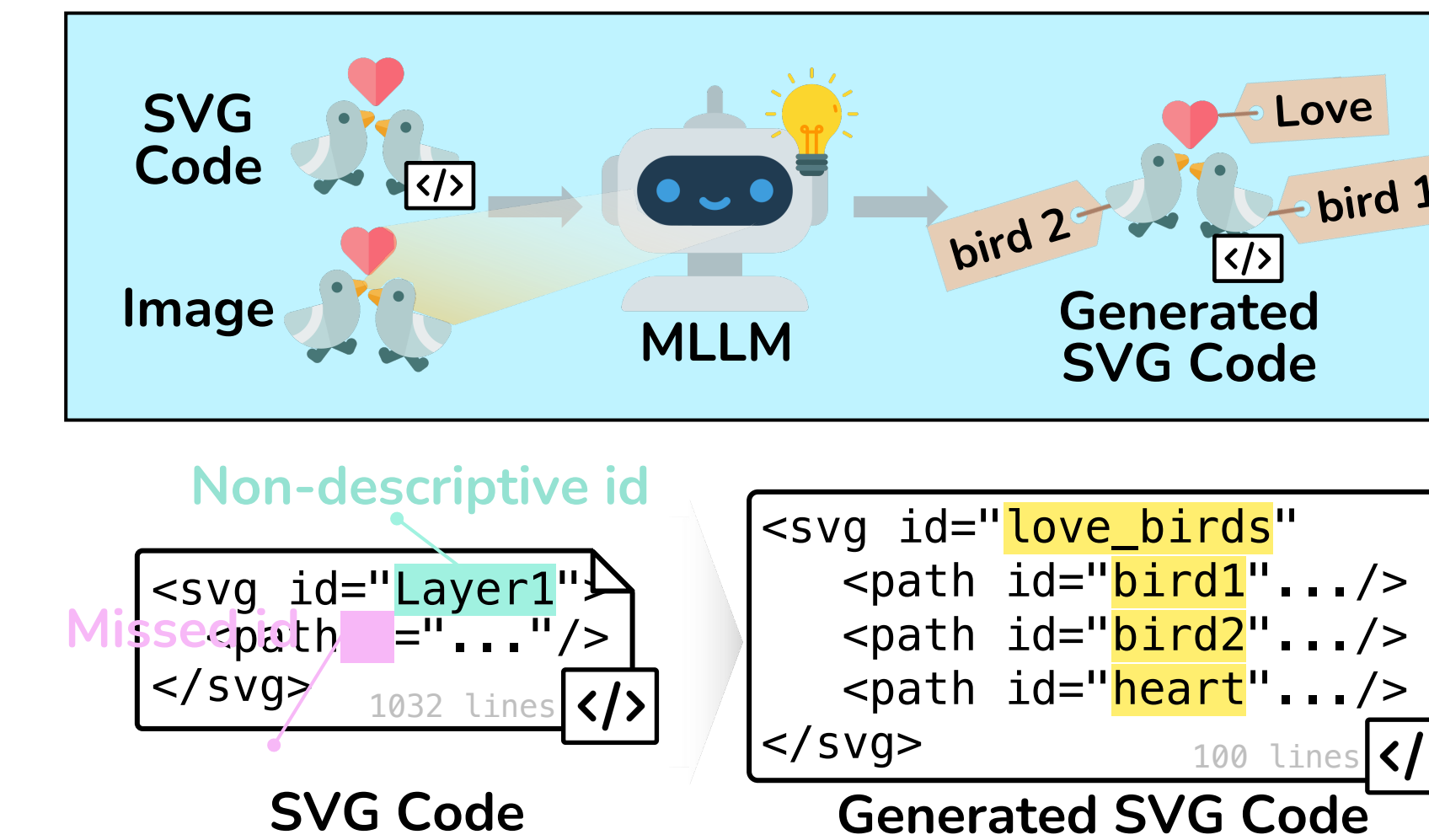
- SVGs provide structure cues, but in a verbose hierarchical form

This `[X]` (instruction) in one word:

- Explicit One-word Limitation (EOL)** instructs a model to summarize an input into a single word, originally proposed for text embeddings.

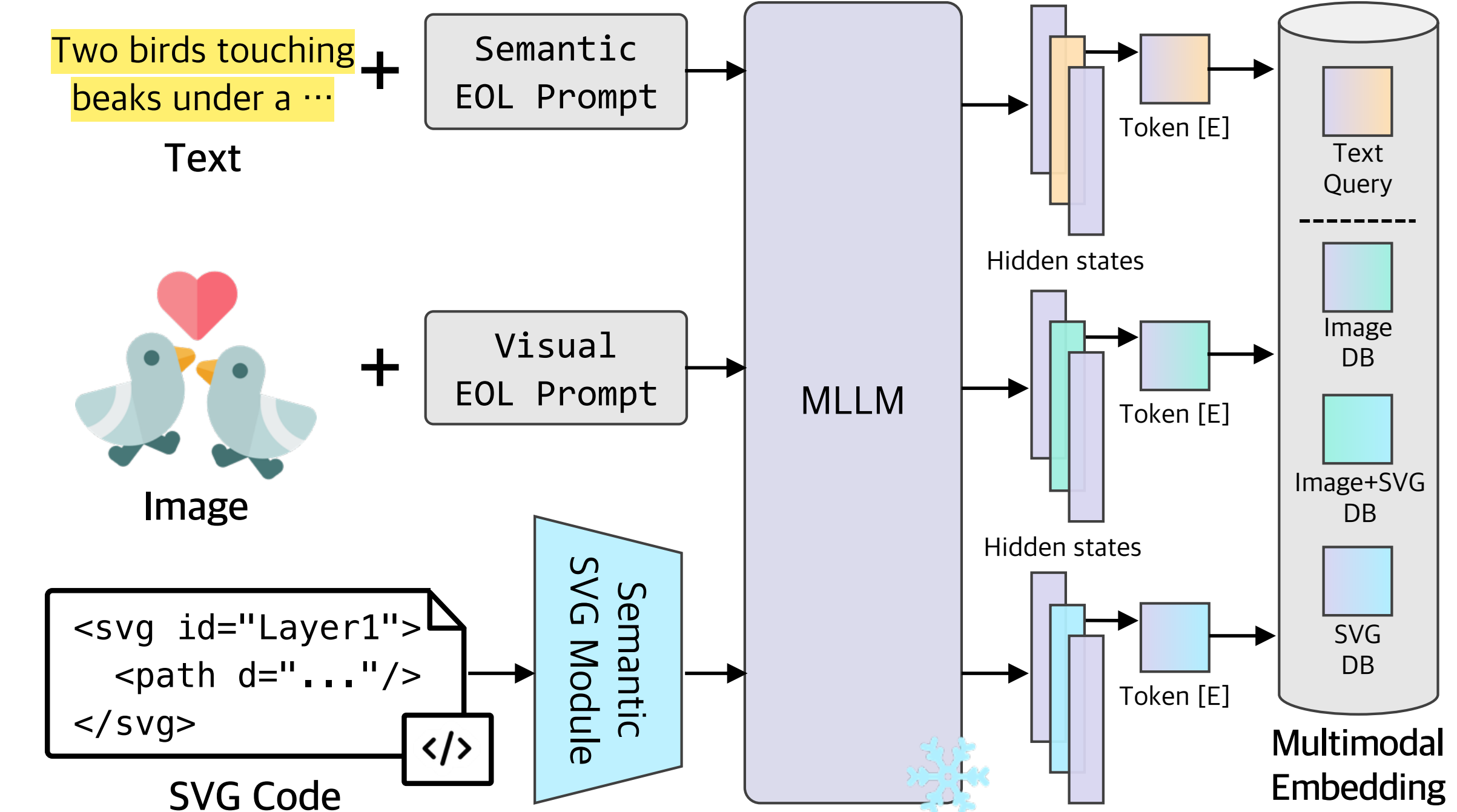
Proposed Method: Multimodal EOL (mEOL) Embedder with Semantic SVG Module

Semantic SVG Module



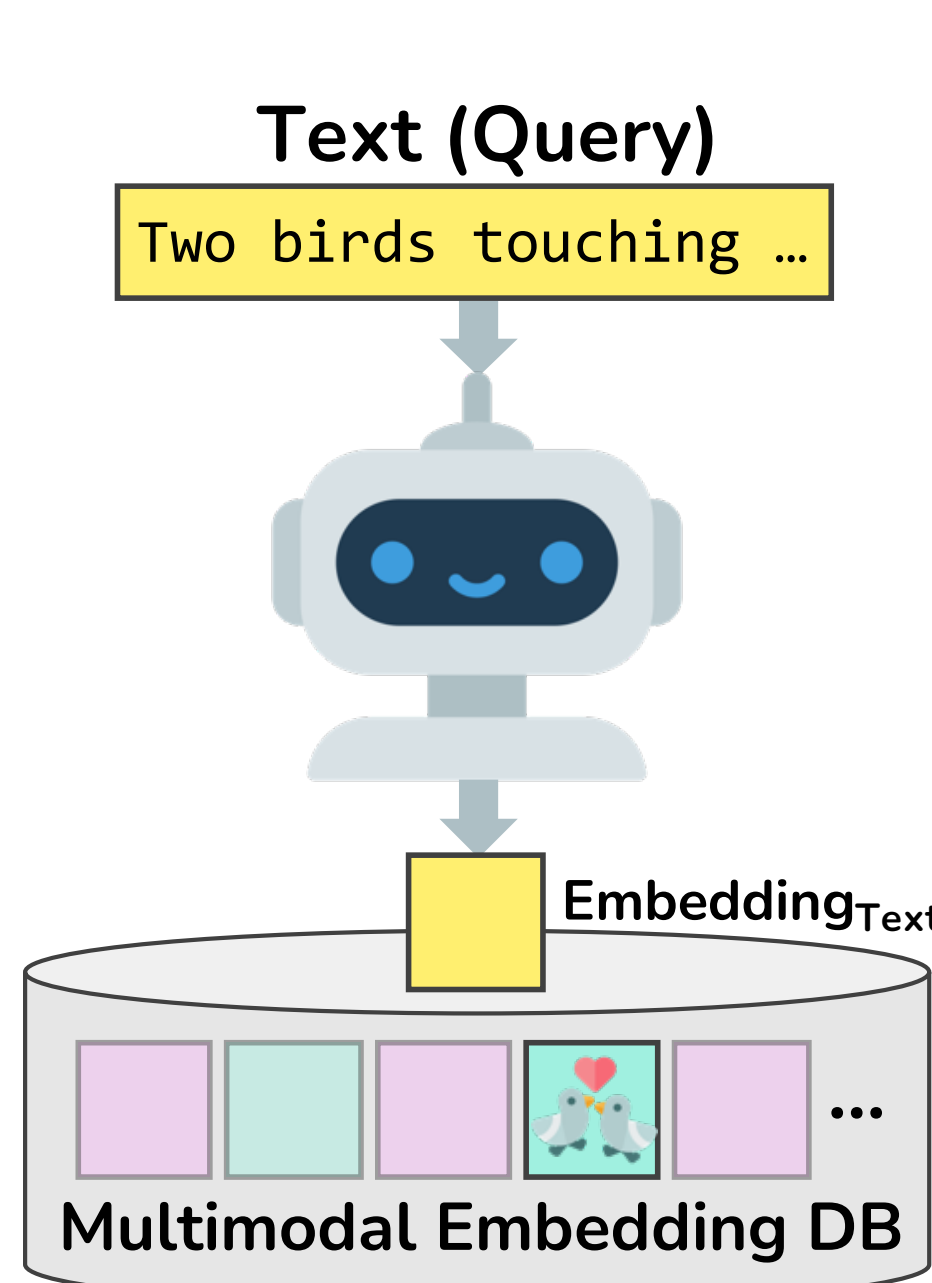
- Assigns meaningful identifiers and simplifies SVG structure via visual reasoning
- Enables structure-aware text-to-SVG retrieval without fine-tuning with mEOL Embedder

Multimodal EOL (mEOL) Embedder without Fine-Tuning



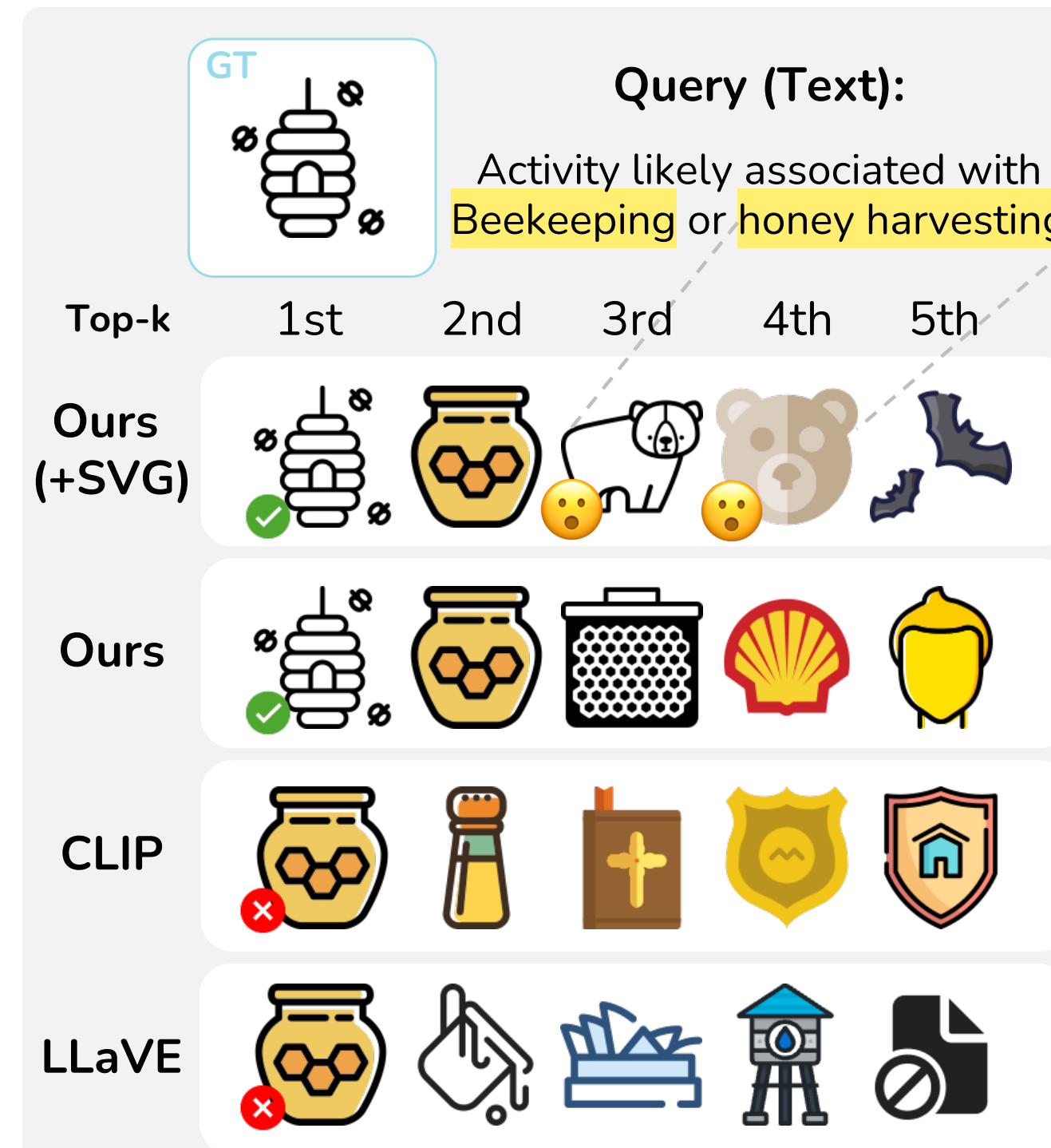
Experimental Setting and Cross-Modal Retrieval Results

Cross-modal retrieval (Text to Image / SVG)



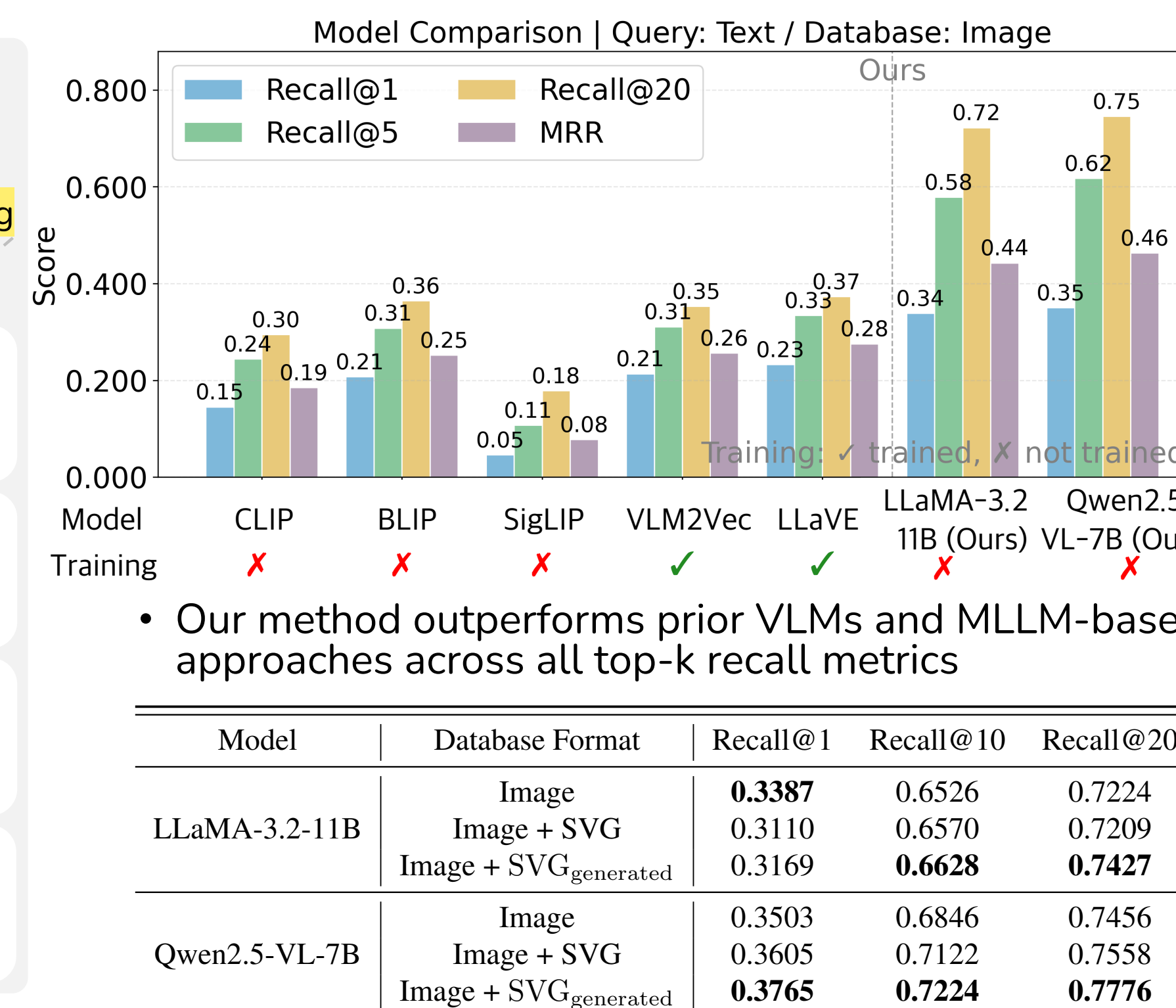
- Cross-modal retrieval with fixed text queries and varying database modalities

Qualitative Results



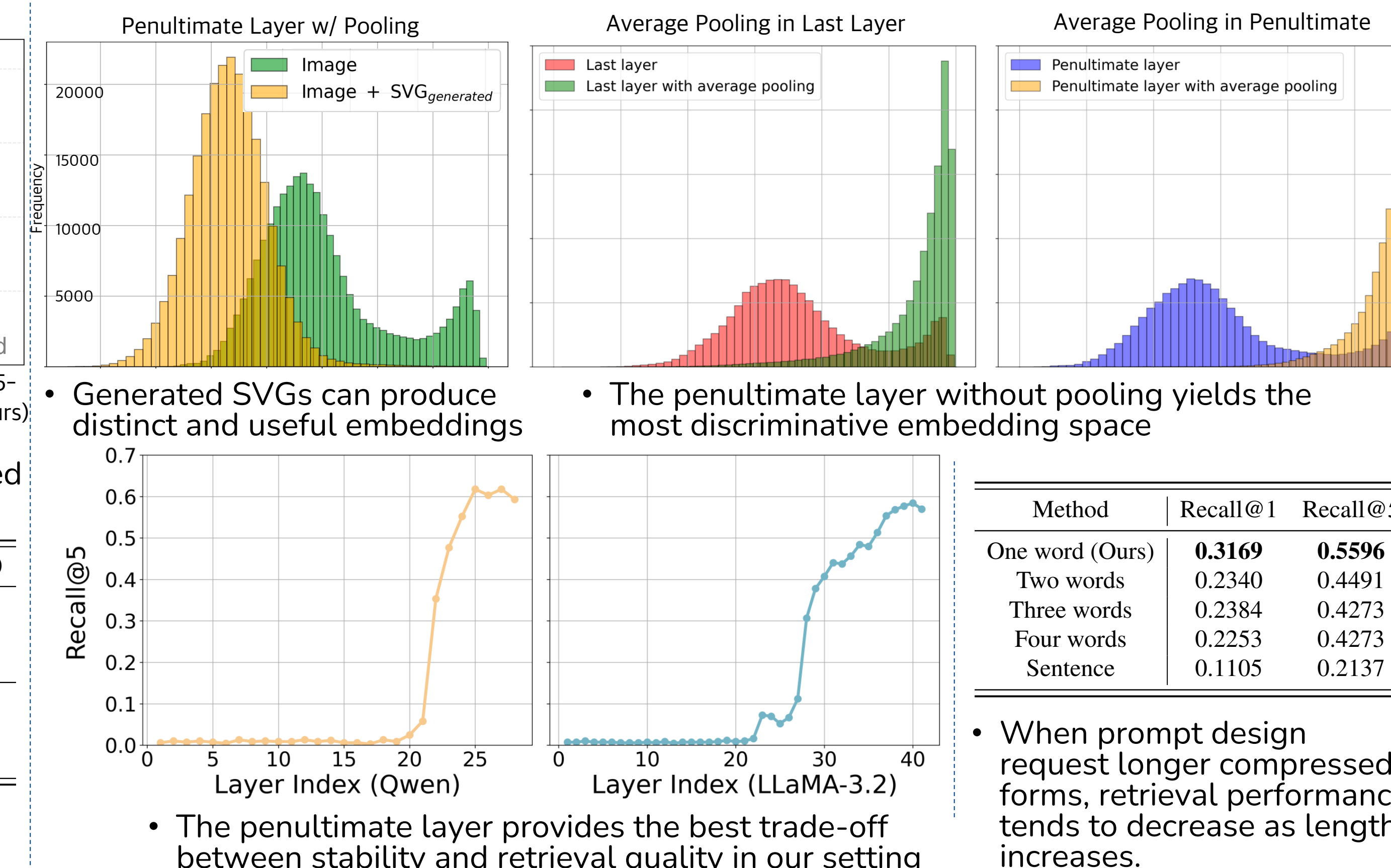
- Structure-aware retrieval captures deeper semantic associations

Main Results



- Generated SVG improves retrieval performance beyond Image+raw SVG fusion

Ablation Studies



Takeaways

- Structure is a hidden cue:** SVG structure provides useful signals for multimodal alignment
- One token is enough:** A single-token embedding suffices for cross-modal retrieval
- Semantic Module makes SVG embedding-friendly:** Raw SVG code is rewritten into compact, interpretable structure
- First text-to-SVG retrieval:** We introduce and evaluate text-to-SVG retrieval
- Instruction-level control without fine-tuning:** Competitive SVG retrieval is achieved without additional training